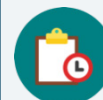


1.3 Interpreting Function Notation in the Real World

Lesson Introduction

 Benchmark	MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.		 Learning Target	I can evaluate a function for a specific input value of the domain and interpret it in a real-world context.
 Benchmark Coherence	Building On MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.		Working Toward MA.912.F.3.3 Solve mathematical and real-world problems involving functions that have been combined using arithmetic operations.	
 Essential Questions	- How can evaluating a function be mathematically accurate but not viable for a context? - What impact does the context of a function have on the reasonableness of the answer?			
 Lesson Narrative	In this lesson, students will continue with evaluating functions, adding in a new layer of interpretation. Students will work with functions that represent real-world scenarios to interpret values and assess their reasonableness based on the context.			
 Component Summary	1.3.1 Warm-Up (~5 min.)	Fill-in-the-blank activity to construct two functions that are both true (<i>SE p. 16</i>)	1.3.4 Try It (5 min.)	Evaluate and interpret a function that represents a real-world context (<i>SE p. 21</i>)
	1.3.2 Guided Instruction (15 min.)	Interpreting functions presented in multiple forms (<i>SE p. 17</i>)	1.3.5 Your Turn! (5 min.)	Evaluating and interpreting functions that represent a real-world scenario independently (<i>SE p. 21</i>)
	1.3.3 Guided Practice (15 min.)	Interpreting real world functions (<i>SE p. 19</i>)	1.3.6 Wrap-Up (~5 min.)	Self-assessment of interpreting functions (<i>SE p. 22</i>)
 Resources	Glossary		Materials	
	Key Terms: N/A Additional Terms: - Function - Input - Dependent		N/A	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider		
	- Students may not fully understand context to interpret it appropriately. - Students may evaluate function and forget to interpret if the value is reasonable. - Students may make calculation errors when evaluating functions.	- This lesson requires a lot of reading as each problem involves a real-world context. Consider leveraging word problem strategies (see Unit Overview for an example) that students can use to better understand the problems. - Interpreting real-world contexts is an expectation of many of the benchmarks in Algebra 1. Students may struggle with an interpretation because their experience with mathematics is to arrive at an answer but not to relate that to the context provided. Allow for students to have productive struggles while working through the contexts. Avoid establishing a strategy that may ultimately fail a student as the contexts and their mathematical representations change. Encourage students to share their thinking so that other students can consider different ways of looking at the contexts. - If time is an issue, then consider assigning 1.3.5 Your Turn! for homework.		
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Three Reads Each activity in the lesson (excluding 1.3.1 Warm-Up and 1.3.6 Wrap-Up) involve word problems that impact the work students will complete. Consider leveraging the “three reads” protocol. On the first read, students have a goal of understanding the context. On the second read, students have a goal of understanding the mathematics required. On the third read, the students generate a plan for solving the problem.		MA.K12.MTR.6.1: Reasonableness The core activities of this lesson allow students to learn and practice evaluating functions and interpreting values within the context provided. They will encounter values that are mathematically appropriate but are not reasonable for the context provided. Students will engage in assessing whether values are reasonable throughout the lesson.	
 Differentiate Instruction	1.3.4 Try It provides students with a structured practice opportunity. Student performance during this activity can determine whether they may need additional support during 1.3.5 Your Turn! when they practice individually. In this lesson, students will be asked to fill in the blank to describe the x – and y – axis. Consider having a blank graph available and having students label the graph to visually connect the description to the graph. 1.3.5 Your Turn! provides a valuable formative assessment opportunity to determine what students know about the content of this lesson. Students who perform poorly on this activity would need additional support for this content.			
Lesson Notes:				

1.3.1 Warm-Up: Equivalent Function Values

Component Overview: Students will complete a fill-in-the-blank activity to construct two functions that are both true, with digit use limitations. This activity is designed to be done independently.



1.3.1 Warm-Up: Equivalent Function Values

- Fill in the boxes to construct two functions where the final equality statement is true. Use the digits 1 to 9 once each at most.

$$g(x) = \boxed{}x - \boxed{}$$

$$g(\boxed{}) = 26$$

Answers will vary. Examples include:

$$g(x) = 9x - 1; g(3)$$

$$g(x) = 7x - 2; g(4)$$



Students may struggle with the task, as it will be their first time with this type of activity in the course. Consider spending extra time introducing what the expectations for this type of problem are, with particular emphasis on the digit limitations, as there will be many more of these problem types throughout the units.



Consider showing an example of filling in a box and eliminating that number from being used again. For visual learners, write each digit from 0 to 9 on a common location such as a whiteboard. When a digit is used, cross that number out so that visual learners can connect to the idea of that number no longer being available.



Allow students to share out their answers and determine who has correct answers. Have students share out the method(s) and/or strategies they used to arrive at their answer.

1.3.2 Guided Instruction: Interpreting Functions

Component Overview: Students will receive guided instruction on how to interpret a function presented in multiple forms and determine if the values found are reasonable within the context.



1.3.2 Guided Instruction: Interpreting Functions

Recall that functions have a special notation which allows unique relationships to be expressed in a concise and meaningful way.

The value of a function is dependent on the input. The statement $D(2) = 60$ means that for an input of 2, the value of the function, D , is 60.

Functions can be used to model real-world relationships.

1. If $F(t)$ models the outside temperature in degrees Fahrenheit t hours after 6 a.m., the statement $F(2) = 60$ has a specific meaning within the context.

$F(2) = 60$ can be interpreted as 2 hours after 6 a.m., which is 8 a.m., the outside temperature is 60 degrees Fahrenheit.

2. Physics students drop a ball from the top of a 400 foot high building and model its height above the ground as a function of time using the function $H(t) = 400 - 16t^2$. The height, H , is measured in feet and time, t , is measured in seconds.
 - a. Find the value of $H(0)$.
400
 - b. What does $H(0)$ mean in the context of this problem?
The height of the ball before any time has elapsed, or height of the building.
 - c. Evaluate $H(2)$.
336



To review what was learned in 1.2.4 Guided Instruction (Lesson 2), consider the following questions:

- What does the special notation for a function look like? What are some examples?
- What does it mean when $f(x)$, $g(x)$, $h(x)$, etc.?



Consider using question 1 as a modeling opportunity for thinking through a problem and determining its interpretation. Ask students while working through the problem about whether the interpretation is viable and appropriate, using evidence from the problem. Discuss how the interpretation would differ if the values in the function were reversed.



Student performance with question 2 can provide data for potential differentiation or support. Students who can do question 2 parts a and c, but struggle with b and d understand how to evaluate functions from prior lesson, but don't understand context and interpretation from current lesson.

1.3.2 Guided Instruction: Interpreting Functions (continued)

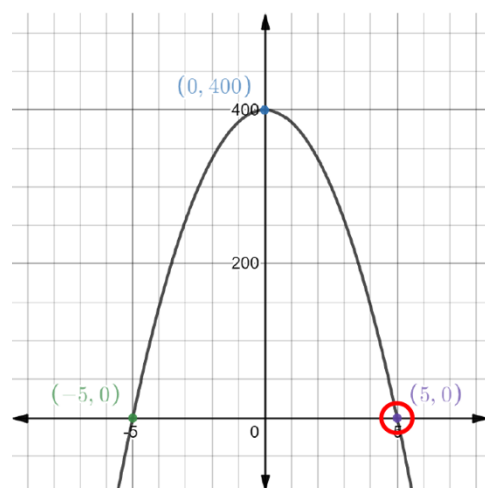
- d. Interpret the meaning of $H(2)$ based on the given context.

After 2 seconds has passed, the ball is 336 feet above ground.

- e. José is asked to interpret the value of $H(-3)$. Is interpreting this value reasonable in the context of this situation?

In this context, a negative value is not reasonable. Since t represents time in seconds, it cannot be negative.

3. The graph of $H(t)$ is shown on the coordinate grid.



- a. Based on the context, the x -axis represents *Time elapsed since dropping the ball* and the y -axis represents *the height of the ball from the ground*.

- b. What will the height of the ball be when it hits the ground?

0 feet

- c. Circle the point where the ball will hit the ground.

Answer on graph



For question 2 part e, consider engaging students in a conversation about reasonableness in this context. This can help reinforce the idea of interpreting the meaning, addressing potential misconceptions, and reinforcing the concept of viability. Ask students if there are any potential situations where the negative value would be reasonable. Be careful about establishing with students that “there is no such thing as negative time”. A negative value that represents time is establishing what may have occurred before. For some contexts this makes sense. For example, a student may consider that $H(-3)$ is when the ball was, three seconds before it was taken to the top of the building. This is a valid interpretation. If a student does not provide this interpretation, then consider including it for students to discuss.



For visual learners, consider taking the information from the blanks in part a of question 3, and having students label the graph as a group.



- Why is the circle only on the x -intercept at $(5,0)$?
- Why would the other x -intercept at $(-5,0)$ not be circled?
- What about point $(0,400)$? Why is it not circled?
- What does it mean in this context?

1.3.2 Guided Instruction: Interpreting Functions (continued)

- d. Write the point where the ball hits the ground using function notation.

$$H(5) = 0$$

- e. How long did it take the ball to hit the ground after it was released?

5 seconds

- f. Discuss with your partner what $H(1.8) > H(3.4)$ means in terms of the context.

The height of the ball at 1.8 seconds is greater than the height of the ball at 3.4 seconds. This is because the height of the ball, from the ground, decreases as the seconds increase because it is falling further and further from the top of the building.



- How can you determine how long it took the ball to hit the ground after it was released?
- How does the solution relate to the information that is present on the graph?



For question 3 part f, consider allowing students to share out what they believe this means. There are possible misconceptions with this question that could be addressed through a facilitated discussion.



For a challenge, have students reverse the inequality symbol and answer the same questions. Does the resulting interpretation make sense? Is it viable?

1.3.3 Guided Practice: Interpreting Functions

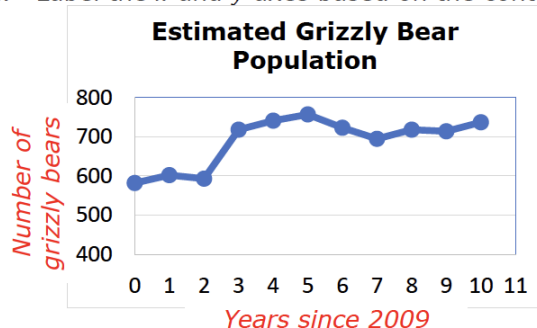
Component Overview: Students will receive a guided practice opportunity to evaluate and interpret real world functions.



1.3.3 Guided Practice: Interpreting Functions

1. The population of grizzly bears at Yellowstone National Park has been threatened with local extinction since 1975. Through many conservation efforts, grizzlies have made a remarkable recovery. The function $y = G(t)$ is modeled below, where t is the number of years since 2009 and G is the number of grizzly bears.

- a. Label the x and y axes based on the context.



- b. Find the approximate value from the graph above.

$$G(3) \approx 718$$

- c. In the context of this problem, $G(3)$ means that 3 years after 2009, there were approximately 718 bears in Yellowstone.

- d. Find the approximate value from the graph.

$$G(8\frac{1}{2}) \approx 710$$

- e. What does $G(8\frac{1}{2})$ mean in the context of this problem?

At $8\frac{1}{2}$ years after 2009 there would be approximately 710 grizzly bears.



For question 1, review the context and ask students what they know from the context. Ask students what key details or information from the problem may be necessary. Help students understand the definition of the variables. Ensure students understand that the values along the x -axis represent the years since 2009. Consider these questions:



- What was the population of grizzly bears in Yellowstone in 2009? In 2019?
- What year does it appear that the population of grizzly bears was at its highest?
- Approximate the population. (~750).



Students may get stuck in the mathematical language in that they lose focus on the real-world context and the language of time. Encourage students to think through their answer to consider what they are saying when they say $G(3)$. Although it is 3 years since 2009, it would be the year 2012. Many students may miss this connection.

1.3.3 Guided Practice: Interpreting Functions (continued)

- f. Explain why there is not a value of t , such that $G(t) = 710.5$.

The value represents the number of grizzly bears. There cannot be a partial grizzly bear, so in this context the value should be a whole number.



- Could evaluating for 710.5 be mathematically possible?
- Why would it not be a value in this scenario?

2. Matthew filled his bathtub, took a bath, and then drained the tub. The function $B(m)$ gives the depth of the water, in inches, m minutes after he began to fill the bathtub.

- a. Explain the meaning of each statement in this context.

$$B(0) = 0$$

At 0 minutes after Matthew began to fill the bathtub, the depth of the water was 0 inches. The tub began with 0 inches of water in it.

$$B(9.5) = B(20)$$

At 9.5 minutes after Matthew began to fill the bathtub, the depth of the water was the same as it was after 20 minutes.

- b. Discuss with your partner why the inequality $B(1) < B(7)$ is likely true in this context.

The depth of the water 1 minute after beginning to fill the bathtub is likely to be less than the depth of water 7 minutes after beginning to fill the bathtub, even if it was already beginning to drain. It is improbable that Matthew would fill the tub, take a bath, and drain it within 7 minutes.



Students may misinterpret the context of question 2 when completing parts a and b. Consider reviewing the context of the problem with students prior to releasing them to work as a method of addressing potential misconceptions ahead of time. Have them reason through the process of water in a bathtub to better have contextual understanding.



For a challenge, have students reverse the inequality symbol and answer the same questions. Would the tub need to be filling or draining for the inequality to be true?

1.3.3 Guided Practice: Interpreting Functions (continued)

3. A biologist is studying the relationship between a tree's diameter and its height. She records the following data for seven different trees using the function $H(d)$, where H represents the height, in feet, and d represents the diameter, in inches.

Diameter (inches)	2	4	6	8	10	12	14
Height (feet)	6	7	10	15	18	22	28

- a. Fill in the blanks in the following sentence to interpret the statement $H(10) = 18$ in this context.

A tree with a diameter of 10 *inches* has a height of 18 *feet*.

- b. Determine the value of $H(14)$.
28 *feet*

- c. Interpret $H(14)$ for the given context.
A tree with a diameter of 14 inches has a height of 28 feet.



Why is it important to connect the labeling of the table with the information from the context?



- *Where on the table did you look to find the value of $H(14)$?*
- *What if the question were written as $H(x) = 14$?*
- *Would the table provide the information needed?*



Allow students to share out not only their answer for question 3 part c, but also to focus on their method for understanding how to interpret the value for the given context. Have students agree or disagree with an interpretation, allowing for student discourse to occur.

1.3.4 Try It: Evaluating and Interpreting Functions

Component Overview: Students will practice being able to evaluate and interpret a function that represents a real-world context. For this activity, release students to work with a partner on completing questions 1 part a and part b. Call the students back together and review the answers as a class.



1.3.4 Try It: Evaluating and Interpreting Functions

1. For groups of 80 or more people, a charter bus company determines the rate per person according to the formula $R(n) = 8 - 0.05(n - 80)$, where n represents the number of people and R represents the rate, in dollars, charged to each customer.

- a. Find each value and interpret it in context.

$$R(75) = 8.25$$

When there are 75 people in the group, the price per customer would be \$8.25. However, we are told that this function is only for groups of 80 people or more so 75 is not a realistic input.

$$R(85) = 7.75$$

When there are 85 people in the group, the price per customer is \$7.75.

- b. Why is $R(75) > R(85)$ in this context?

When the number of people exceeds 80, there is a discount in the price per customer.



Remind students to be careful to review what the information is about, what question is being asked, and what key information can be gathered from the context. Remind students to refer to it when working on interpreting the values and determining reasonableness.



Students may get stumped with the value of $A(75)$, since the problem says for groups 80 or more. Students can evaluate this number but should recognize that the input value itself is not realistic because the company only uses this function when the value is 80 or greater. Do not give this information to students up front but be sure to review this with students at the end of the activity to reinforce the concept of reasonableness.

1.3.5 Your Turn!

Component Overview: Students will independently practice evaluating and interpreting functions that represent a real-world scenario. For this activity, release students to work independently on this activity. Once completed, review answers as a whole group.

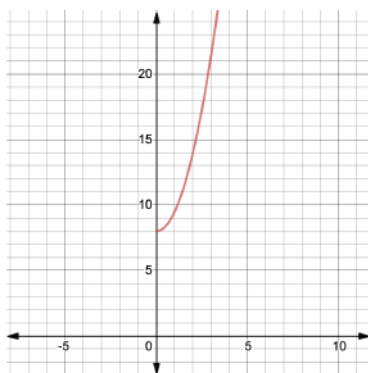


1.3.5 Your Turn!

- The size of a fungal colony is modeled by the equation $z(t) = \frac{3}{2}t^2 + 8$, where z is the size of the colony, in microns, and t is the time, in hours.

A graph of the function $z(t)$ is shown.

Using the graph, approximate the values and interpret their meaning in this context.



- $z(3) = 21$
After 3 hours, the size of the colony is approximately 21 microns.
- $z(0) = 8$
At 0 hours, the size of the colony is 8 microns. The starting size of the colony is 8 microns.
- Why does the graph only display y -values for the function when $t \geq 0$?
In this context, the t -values refer to time. Since time cannot be negative, the corresponding y -values would only appear for t -values that are 0 or greater.



For students who performed poorly on the prior activity, consider using this activity time to work with a small group who may need additional support. Walk them through this problem step-by-step in a gradual release method, consistently checking for understanding.



At the conclusion of this activity, consider having students compare their answers with their partner to determine any potential mistakes that were made. Allow students to share out their answers for question 1 part c to allow for more thoughts and comparisons.



Do you think that a different context would allow for the graph to display y -values for the function when x is less than 0? If yes, what type of context would allow for this to occur?

1.3.6 Wrap-Up: Reflection

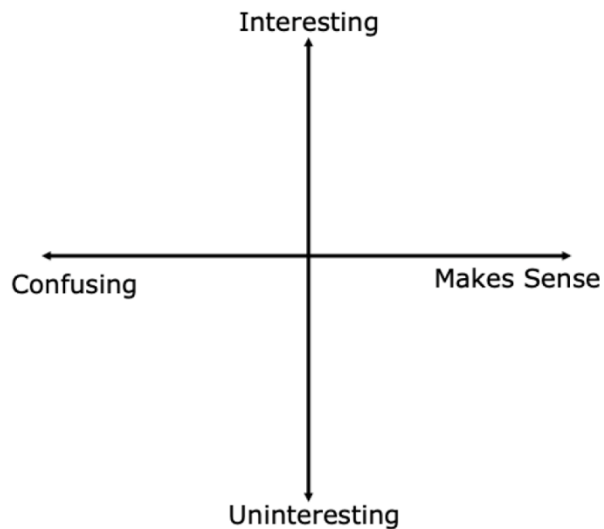
Component Overview: Students will complete a self-assessment of their level of interest and understanding of interpreting functions. This activity should be done independently to allow for accurate student data.



1.3.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you feel you are on today's learning target: "I can evaluate a function for a specific input value of the domain and interpret it in a real-world context."

Answers will vary.



Students may need a demonstration of this to get better data. For example, consider putting this chart on the board with other concepts such as "bread making," and plot a point and use another concept such as "surfing". Highlight for students what their plotted point represents on this graph.